INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Governance, Compliance, and Risk (GCR)

Course 2: GRC102: Information Security Governance

Log Analysis Assignment

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Report on Log Analysis: Global Tech

Executive Summary:

Global Tech is a mid-sized financial technology company. In this report we submit an analysis of security logs from the SIEM system and identified key critical security events and patterns. Based on the analysis done once the data was transformed, we picked the two topmost security issue and provided an explanation on why it is a key concern and suggested a simple, actionable remediation step for each identified issue. We have also provided supporting analysis by giving top issues from the various logs analyzed and supporting screenshots of what the analysis is based on, these are additional detailed resources that can be reviewed to gain a deeper perspective on why the key issues were picked as top.

Key Findings:

After the analysis the following two stood out as the most critical, we share the finding, the reason and the remediation suggested.

a. Unresolved Data Exfiltration Incidents on Multiple Systems

Finding: Several systems, including the DB-Server, Webserver and user workstations have ongoing incidents related to data exfiltration which still remained unresolved with the status either under investigation or detected, this is indicative of a threat that has not yet been fully addressed. **Why it is a concern:** Data Exfiltration is a major security issue because it may lead to sensitive or confidential information loss or theft. There is a suggestive pattern which could be an indication of either active or coordinated attack based on the issue being identified across multiple systems. **Suggested Remediation:** from the above concern, we recommend immediate strengthening of the Data Loss Prevention measures, detailed review of

all affected system and setting up alerts on unusual data transfers or on suspicious outbound traffic.

b. **Firewall System is Under Multiple Types of Attacks**

Finding: The firewall is showing many different types of attack attempts from the given logs, including DDoS, unauthorized access attempts, phishing and possible data exfiltration, some of the incidents are still marked as under investigation or detected(open). **Why it is a concern:** Even though firewalls sit at the network edge and are often targeted, the presence of many different incident types on the same firewall could be an indicator of a serious problem, this might be related to either a misconfiguration, missing update or compromise in the firewall. **Suggested Remediation:** The firewall should be checked and investigated thoroughly which includes, reviewing the rules and policies and confirmation on no sign of compromise. We also suggest updating the firmware and also limiting and monitoring admin access.

Conclusion on Key Findings:

While incidents are being resolved as indicated by the log analysis, other critical issues especially those touching on data exfiltration and attacks on core infrastructure remain active, this could pose serious risks to the organization if not addressed promptly. The remediation steps given are quick wins and basic steps like better monitoring (continuous effective), reviewing systems configurations and closing open incidents (establishing an effective SLA), these can effectively reduce the threat.

Supporting Analysis

1. **Analysis of the Logs from the Vulnerabilities identified:**

From the logs provided, we did an analysis by grouping the vulnerabilities into severity count and the systems affected and analysed the issues not resolved and not under the status Remediated.

Finding 1: Critical Vulnerabilities not Remediated

Based on the figure appendix (Fig1), we notice that we have critical vulnerabilities still have the status “Detected”, which means they have not been fixed yet. Critical issues can translate to high impact and possibility like an attacker getting access to systems, this is something that must be addressed as soon as possible. **Suggested Action:** Patching systems quickly and ensuring we have process in place to ensure priority is given to critical issues and fixing as soon as identified.

Finding 2: High Severity Issues on App-Server-03

Hight severity vulnerabilities pose a big risk, leaving them unresolved could lead to problems if an attacker finds ways to exploit it. We have noted several high vulnerabilities in the app server based on the Fig1(refer to appendix). **Suggested Actions:** The team to review how systems are maintained and update regularly. Regular scans are also recommended.

Finding 3: Many False Positives Across the Logs

Based on Fig2(Refer to appendix) there were multiple entries in false positives (15), including a few labelled High and even critical. This might show that the tool used to scan for vulnerabilities is flagging issues that are not actually real. Too many false alarms might translate to wasted time and effort and might lead to missing real issues or disregard of a critical issue. **Suggested Action:** Adjusting or tuning the security tool to reduce the false alert and regularly checking and updating how scans are set up.

2. **Authentication Failure Log**

From the data Given, please refer to appendix 3, an analysis of the user and the authentication error was done, reviewing the number of times the user account was logged.

Finding1: Inconsistent Lock out Policy

From the data, we can see that despite the admin account having 20 failed logins it got locked out only 3 times while sysuser had 8 lockouts for 25 attempts. This could be an indication that lockout policies are not standardized, giving attackers more chances on some accounts than others. **Suggested Action:** Review and Standardize lockout rules for all users, especially admin-level accounts.

Finding 2: The guest Account Heavily Targeted

There were target attempts (22 attempts, 7 lockout) which might mean an attacker is testing default or unused accounts hoping to find one that is still active. Even if a guest has limited permissions, it could still be misused if not well secured. **Suggested Remediation:** Disable or remove guest account if it is not needed, if needed monitor closely for security issues.

3. **Security Incidents Logs**

An analysis of the data provided to find the pattern in the type of incidents, resolution status and targeted system assisted in finding critical issues as show:

Finding 1: Multiple Systems Experiencing Serious, Ongoing Security Incidents.

Systems like the Firewall and DB server and webserver are dealing with variety of unresolved security problems including Data exfilterration, phishing and DDoS attacks and unauthorized access. The firewall, usually for protection of the network is facing many kinds of attacks, this could be an indicator of firewall misconfiguration. **Suggested Action:** Review and Tighten security on affected systems, implement stronger monitoring and DLP for systems with exfiltration issues.

Finding 2: High Number of Unresolved Incidents

Looking at the analysis, about 16 incidents have been closed, however 14 are still open(detected, investigation), this indicated that even though the organization is handling some issues, many serious problems remain unresolved. A deeper look shows that incidents like data exfiltration and unauthorized access which could be potential breaches or system compromise, are not yet fully fixed. This could leave the company exposed to attacks longer than necessary. Which necessitates speeding up the process. **Suggested Action:** Improve incident response by assigning clear roles and following up on unresolved cases.

Appendix

Status	Detected

Count of Vulnerability Information	Column Labels				
Row Labels	Critical	High	Low	Medium	Grand Total
System app-server-03		3	2	1	6
System db-server-02	1	1	1	2	5
System firewall-01		1		1	2
System webserver-01	3	1	1	2	7
Grand Total	**4**	**6**	**4**	**6**	**20**

Fig1: Analysis of Systems vs count Identified Vulnerabilities

Status	False Positive

Count of Vulnerability Information	Column Labels				
Row Labels	Critical	High	Low	Medium	Grand Total
System app-server-03		2	1		3
System db-server-02		1	1	1	3
System firewall-01		3	1	1	5
System webserver-01	1	1		2	4
Grand Total	**1**	**7**	**3**	**4**	**15**

Fig2: False Positive

Count of Date and Time	Column Labels			
Row Labels	Account locked	Invalid password	User not found	Grand Total
admin	3	8	9	20
guest	7	8	7	22
jdoe	5	5	6	16
network_admin	4	5	8	17
sysuser	8	8	9	25
Grand Total	**27**	**34**	**39**	**100**

Fig3: Authentication Analysis

Status	(All)

Count of Date and Time	Column Labels					
Row Labels	Data Exfiltration	DDoS Attack	Malware Infection	Phishing Attempt	Unauthorized Access	Grand Total
db-server-02	2	2	1	1	1	7
firewall-01	2	2	1	1	2	8
user-workstation-10	2		3	1	1	7
webserver-01	3		2	3		8
Grand Total	**9**	**4**	**7**	**6**	**4**	**30**

Fig 4: All Incidents Logged

Status	(Multiple Items)

Count of Date and Time	Column Labels					
Row Labels	Data Exfiltration	DDoS Attack	Malware Infection	Phishing Attempt	Unauthorized Access	Grand Total
db-server-02	1				1	2
firewall-01	1	2		1	1	5
user-workstation-10	2		1	1		4
webserver-01	1		1	1		3
Grand Total	**5**	**2**	**2**	**3**	**2**	**14**

Fig 5: All Incidents not resolved(Detected, Investigating)